

Explanation Scoring Rubric

Study: Evaluating AI Guardrails and Their Effect on Students' Independent Critical Thinking

Purpose

This rubric is used to evaluate the **quality of student explanations** in problem-solving tasks.

The goal is to measure **depth of understanding**, not just correctness of the final answer.

Scoring Criteria

Score	Description
3	Correct answer with clear, logical, and well-structured explanation . Demonstrates strong understanding and reasoning.
2	Correct answer but partial or weak explanation . Shows some understanding but lacks clarity or depth.
1	Incorrect answer but attempted explanation . Indicates partial understanding or effort to reason.
0	No answer or no meaningful explanation provided .

Evaluation Guidelines

When assigning scores, consider:

- **Clarity** → Is the explanation easy to understand?
 - **Correctness** → Is the reasoning logically sound?
 - **Depth** → Does the student explain *why*, not just *what*?
 - **Independence** → Is the explanation in the student's own words?
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Examples

Score 3 (High Quality)

The model is overfitting because it performs well on training data but poorly on unseen data. This happens because the model memorizes patterns instead of learning generalizable features. Techniques like regularization or cross-validation can help reduce overfitting.

Score 2 (Moderate Quality)

The model is overfitting. It works well on training but not test data. We can fix it by improving the model.

Score 1 (Low Quality)

The model is wrong because accuracy is low. Maybe we need more data.

Score 0 (No Response)

(No explanation provided)

Usage in Study

- Each response was scored using this rubric.
- Scores were used to compare explanation quality across groups.
- Higher scores indicate **better reasoning and deeper understanding**.